A Project Report on

DEEP EMBEDDED CLUSTERING-BASED SEMANTIC USER GROUPING FOR UAV-ASSISTED DISASTER COMMUNICATION

submitted in partial fulfilment of the requirements for the course

MACHINE LEARNING
25SC2107E

In

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by

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DEPARTMENT OF **ELECTRONICS & COMMUNICATION ENGINEERING**

Name of Title: Deep Embedded Clustering-Based Semantic User Grouping for UAV-Assisted Disaster Communication

1. Abstract:

Deep Embedded Clustering (DEC) for Semantic User Grouping in UAV-Assisted Disaster Communication addresses the challenge of maintaining efficient communication in disaster-affected areas where terrestrial communication infrastructure may be damaged or unavailable. UAVs can provide temporary wireless connectivity to affected users, but efficient communication becomes difficult when a large number of users have different locations, channel conditions, emergency priorities, data requirements, and delay constraints. Conventional clustering approaches that mainly rely on similarity or spatial characteristics may not adequately represent the diverse communication requirements of disaster users. Therefore, this project proposes a semantic user grouping approach based on Deep Embedded Clustering (DEC) to identify groups of users with similar communication and emergency requirements. The proposed approach considers six input factors: user location, channel condition, emergency priority, requested data type, delay requirement, and semantic similarity. DEC is used to learn meaningful representations of the user information and organize users into clusters according to their combined characteristics. By incorporating both communication-related and semantic information, the proposed approach aims to form more meaningful user groups for UAV-assisted disaster communication and support efficient utilization of available communication resources.

The proposed system will be implemented using Python and suitable machine learning libraries for data preprocessing, model development, clustering, visualization, and evaluation. User data will be prepared using the six selected factors and processed before applying the DEC model. To assess the effectiveness of the proposed clustering approach, its results will be compared with conventional clustering methods, specifically K-Means and DBSCAN. The quality of the generated user groups will be evaluated using the Silhouette Score, Davies–Bouldin Index, and Calinski–Harabasz Score. These metrics will provide different measures of cluster separation, compactness, and overall clustering quality without assuming predefined class labels. The experiments will be conducted using software-based simulations with suitable user distributions and communication conditions representing disaster scenarios. The comparison will help determine whether DEC can provide more meaningful semantic grouping than the selected baseline methods when multiple user characteristics are considered simultaneously. The proposed framework is expected to support better organization of disaster-area users and provide a foundation for more efficient UAV-assisted communication and resource management. The project combines machine learning and advanced communication concepts to address a practical communication challenge in emergency scenarios and can be further extended through additional datasets, features, clustering techniques, and communication optimization methods.

2. SOFTWARE REQUIREMENTS:

Python 3.x
Google Colab / Jupyter Notebook
NumPy
Pandas
Scikit-learn
TensorFlow / Keras
Matplotlib
Seaborn

Signature of the Supervisor

Course Instructor